MODULE 4

Natural Language Processing (NLP)

- Natural Language Processing (NLP) is a branch of Artificial Intelligence (AI) that focuses on the interaction between computers and human language.
- The main goal of NLP is to enable machines to read, understand, interpret, and generate human language effectively.
- NLP combines concepts from linguistics, computer science, and machine learning.
- It helps machines to process human communication in a way that is meaningful and useful..
- NLP techniques are crucial for building systems that can interact naturally with humans.
- **Python provides libraries for NLP tasks**, and the most commonly used library is **NLTK (Natural Language Toolkit)**.

Key Terms in NLP

- **Phonology** – Study of sounds in language.
  - *Example:* "bat" vs "pat" differ by a single sound.
- **Morphology** – Study of word formation and structure.
  - *Example:* "running" → root = run, suffix = ing
- **Morpheme** – Smallest unit of meaning in a language.
  - *Example:* "unhappy" → "un" (not) + "happy"
- **Syntax** – Rules that govern sentence structure.
  - *Example:* English sentence: Subject + Verb + Object → "She (S) reads (V) a book (O)."

- **Semantics** – Meaning of words and sentences.

Example: "I saw the man with a telescope" → could mean **I used a telescope** or **the man had a telescope**

- **Pragmatics** – Study of meaning in context.

Example: "Can you pass the salt?" → request, not question

- **Discourse** – Understanding meaning across multiple sentences.

Example: Resolving pronouns: "John went to the market. He bought apples." → "He" = John

- **World Knowledge** – General knowledge used to interpret sentences.

Example: "The cat chased the mouse." → Knowledge that cats hunt mice

- **Tokenization** – It is a technique in NLP that involves splitting a sentence or phrase into smaller units known as tokens.

Example: "I love NLP." → ["I", "love", "NLP", "."]

- **Part-to-Speech Tagging** – This technique in NLP involves the process of identifying and labeling works in a sentence based in their part of speech (noun, verb, adjective).

Example: "She runs fast." → She/PRON, runs/VERB, fast/ADV

Steps in NLP

5 Steps in Natural Language Processin

1. Lexical Analysis
2. Syntactic Analysis
3. Semantic Analysis
4. Discourse Integration
5. Pragmatic Analysis

1. Lexical Analysis

- **Definition:** Lexical Analysis is the process of analyzing text at the **word level**, identifying basic linguistic units, and preparing it for further processing.

Components of Lexical Analysis:

- **Tokenization**
  - **Definition:** Breaking text into smaller units like words, subwords, or sentences.
  - **Example:** "I love NLP." → ["I", "love", "NLP", "."]
- **Stemming**
  - **Definition:** Reducing words to their root or base form, which may not be a valid dictionary word.
  - **Example:** "running", "runner" → "run"

- **Lemmatization**
  - **Definition:** Reducing words to their **dictionary or base form**, considering context.
  - **Example:** “better” → “good”; “running” → “run”
- **Morphological Analysis**
  - **Definition:** Analyzing the **structure and formation of words**, including prefixes, suffixes, and roots.
  - **Example:** “unhappiness” → “un” (prefix) + “happy” (root) + “ness” (suffix)

2. Syntactic Analysis (Parsing)

- **Definition:** Examining the **grammatical structure** of a sentence.
- **Purpose:** Identifies the relationships between words and ensures sentences are grammatically correct.
- **Includes:** Top-down parsing, bottom-up parsing, dependency parsing, etc.

3. Semantic Analysis

- **Definition:** Understanding the **meaning of words, phrases, and sentences**.
- **Tasks:**
  - Word Sense Disambiguation (resolving ambiguous words)
  - Semantic Role Labeling (identifying agent, action, object)
- **Example:** "I saw the man with a telescope." → Determine who has the telescope

4. Discourse Integration

- **Definition:** Understanding **meaning across multiple sentences**.
- **Purpose:** Maintains coherence and resolves references across sentences.
- **Example:** "John bought a car. It is red." → "It" = John's car

5. Pragmatic Analysis

- **Definition:** Understanding intended meaning in **context**, considering social and situational cues.
- **Purpose:** Interprets implied meaning, sarcasm, or indirect requests.
- **Example:** "Can you pass the salt?" → Interpreted as a request, not a literal question

THANK YOU